Bayesian Inference

INFO 521 · Module 4 · Lecture A — Priors, Posteriors & the Beta-Binomial

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Adapted with permission from lecture materials by Clayton Morrison, University of Arizona.

Learning outcomes

By the end of Lecture A you should be able to:

1. **State Bayes' rule for parameters** and name every term.
2. **Choose and read a Beta prior** for a proportion.
3. **Perform the conjugate update** — Beta(a, b) + binomial data → Beta$(a+n_1,\ b+n_0)$ — by hand.
4. **Explain what a posterior buys** that a point estimate cannot.

From “how uncertain?” to “what do we believe?”

Module 3 quantified uncertainty by imagining **repeated datasets**: $\hat{\mathbf{w}}$ scatters with $\mathrm{Cov}[\hat{\mathbf{w}}]$, but the true parameter stays a *fixed unknown*.

Bayes flips the object of study: keep the **one dataset you actually have**, and treat the parameter itself as **uncertain** — a probability distribution over what it might be, updated by evidence.

Frequentist: the *estimate* is random. Bayesian: the *belief about the parameter* is the distribution. Both use the same likelihood.

Bayes' rule, term by term

For a parameter θ and observed data $\mathcal{D}$:

$$\underbrace{p(\theta \mid \mathcal{D})}_{\text{posterior}} = \frac{\overbrace{p(\mathcal{D} \mid \theta)}^{\text{likelihood}}\ \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(\mathcal{D})}_{\text{evidence}}}$$

- **Prior** — belief before this dataset.
- **Likelihood** — the M3 object, unchanged: how probable is the data at each θ.
- **Evidence** — a normalizing constant (for today); it returns in Lecture B.
- **Posterior** — belief after: $\text{posterior} \propto \text{likelihood} \times \text{prior}$.

The running question: hypertension prevalence

Screen adults one at a time and record hypertension (ACC/AHA rule: SBP ≥ 130 **or** DBP ≥ 80 — measured BP only, the leakage-safe Project-2 label).

- Each patient: hypertensive (1) or not (0) — a **Bernoulli**(θ) draw.
- After N patients with n_1 positives ($n_0 = N - n_1$), the likelihood is **binomial** in shape:

$$p(\mathcal{D} \mid \theta) \;\propto\; \theta^{\,n_1}(1-\theta)^{\,n_0}$$

The full NHANES answer (all $5{,}102$ adults): $\theta \approx 0.388$. Today we watch belief *converge* to it patient by patient.

The Beta prior

A prior for a proportion must live on $[0, 1]$. The **Beta** family:

$$p(\theta) = \mathrm{Beta}(\theta \mid a, b) \;\propto\; \theta^{\,a-1}(1-\theta)^{\,b-1}$$

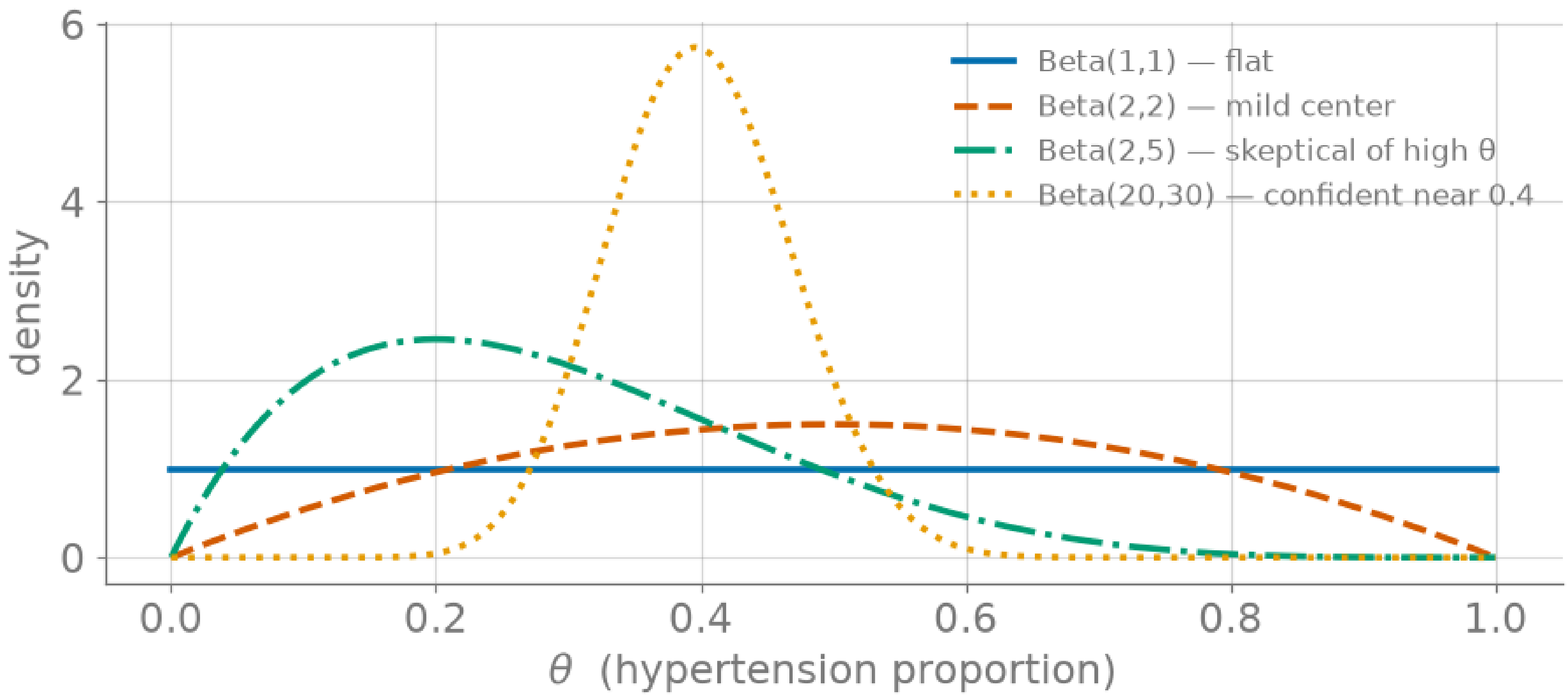

Figure 1: The Beta family (illustrative shapes): a and b act like prior pseudo-counts of 1s and 0s.

Conjugacy: the update is just addition

Multiply prior by likelihood — the exponents **add**:

$$\underbrace{\theta^{a-1}(1-\theta)^{b-1}}_{\text{prior}} \times \underbrace{\theta^{n_1}(1-\theta)^{n_0}}_{\text{likelihood}} \;\propto\; \theta^{(a+n_1)-1}(1-\theta)^{(b+n_0)-1}$$

$$\boxed{\mathrm{Beta}(a,b) \xrightarrow{n_1 \text{ ones, } n_0 \text{ zeros}} \mathrm{Beta}(a+n_1,\ b+n_0)}$$

A prior family that reproduces itself through the likelihood is called **conjugate** — the posterior stays in-family, so updating is exact arithmetic.

Count successes, add to a; count failures, add to b. That's the entire computation.

Watching belief converge

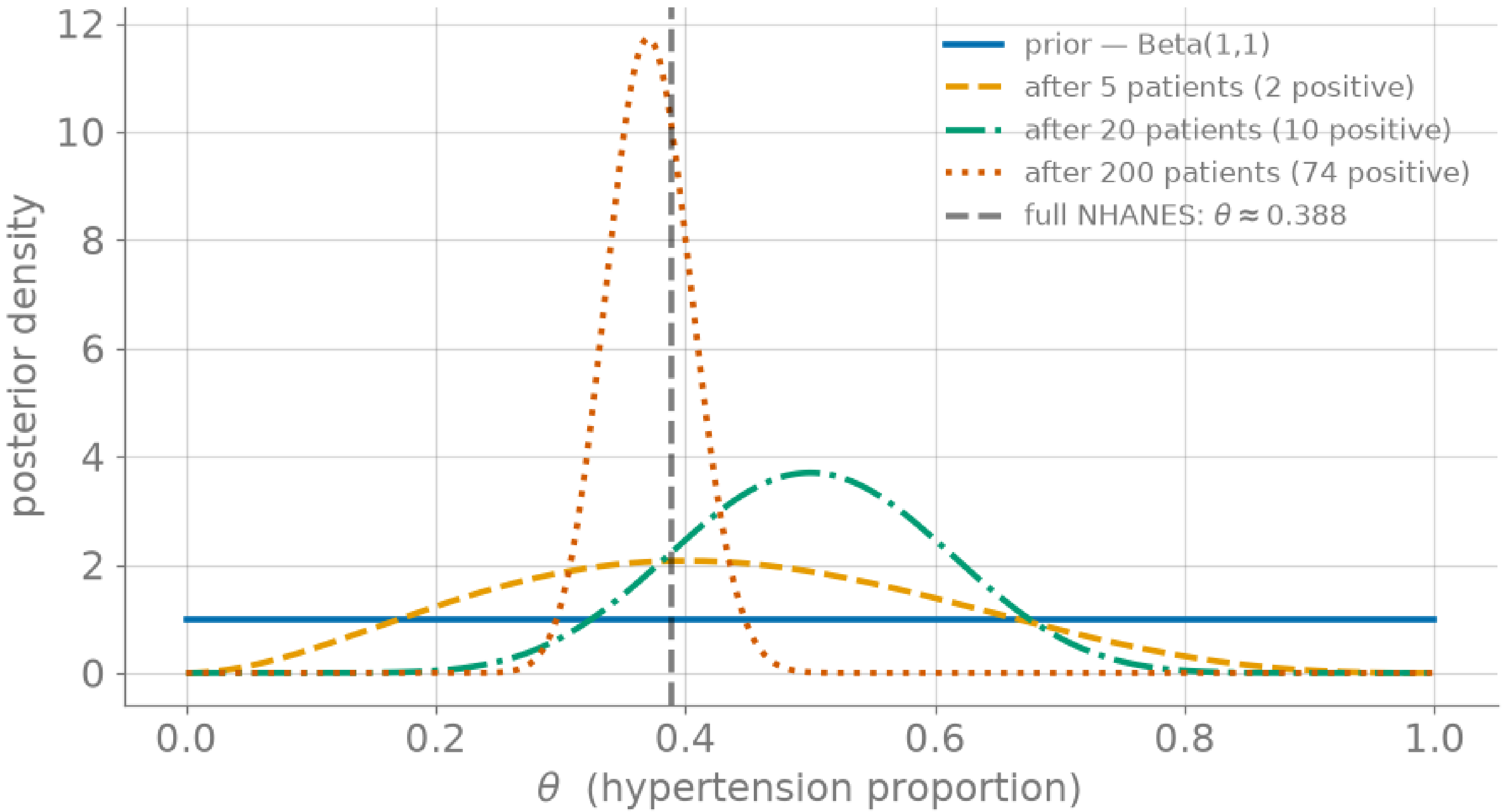

Figure 2: Beta(1,1) updated with real NHANES screening draws (seed 521): 5 → 20 → 200 patients; dashes mark the full-sample θ ≈ 0.388.

Does the prior matter?

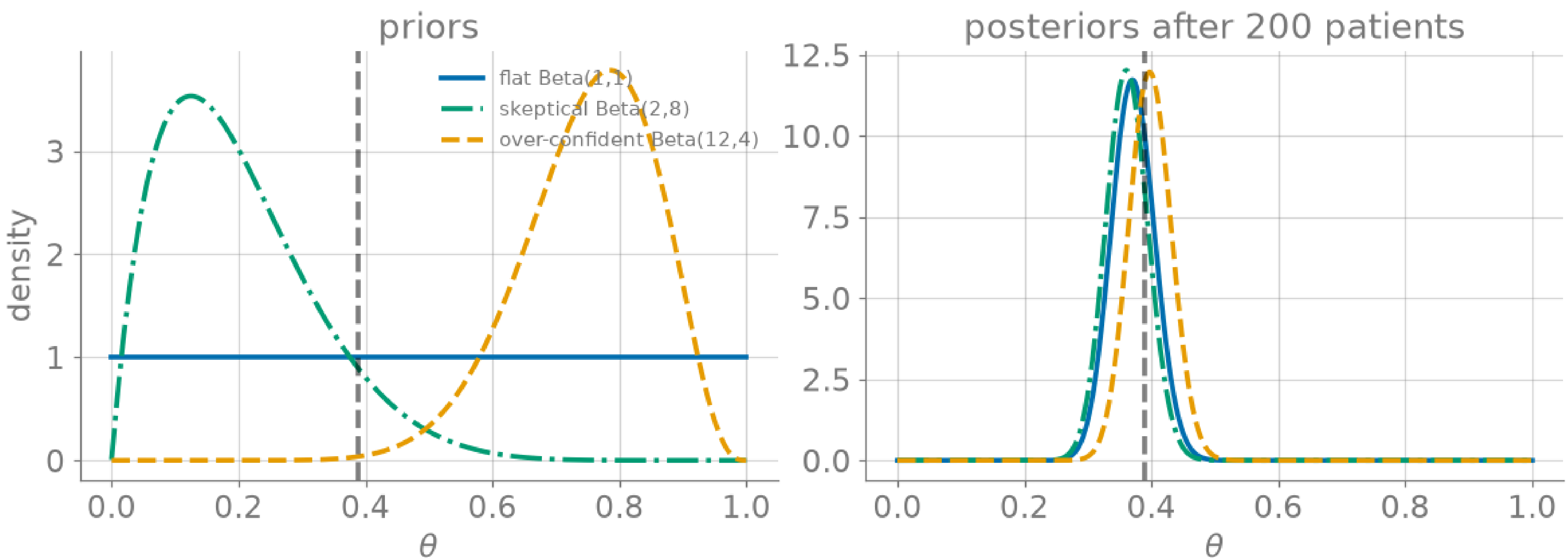

Figure 3: Three priors (left) vs. their posteriors after 200 real patients (right): with enough data, the likelihood dominates.

Live demo: the Bayesian-updating explorer

Set a prior, feed it patients one at a time, and watch the posterior move — the tool runs this lecture’s exact model.

Explorer — static fallback

The iframe above does not print or export to PDF; this snapshot stands in.

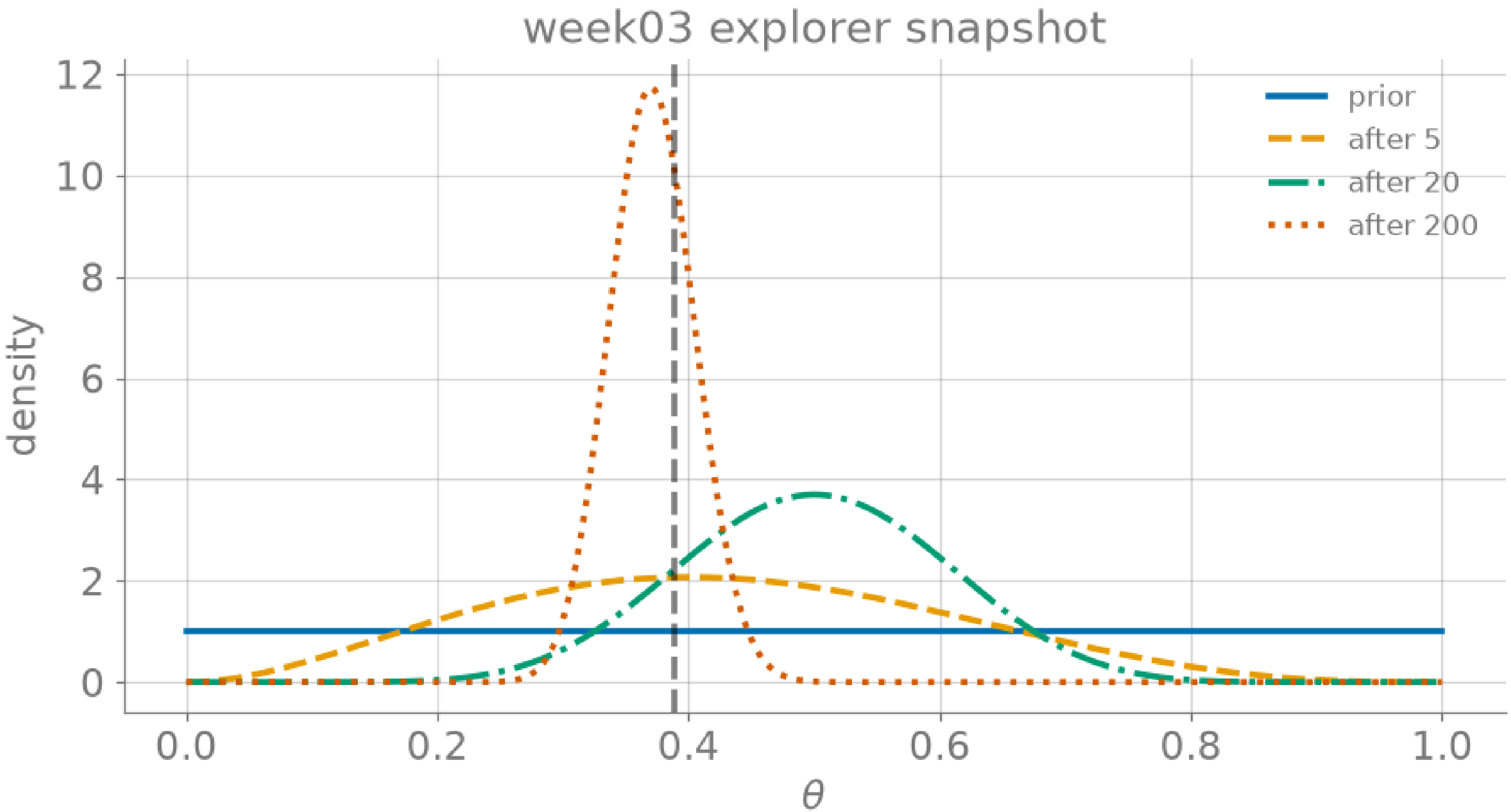

Figure 4: Static fallback: sequential Beta updating (used in print / PDF).

What the posterior buys

The posterior $\mathrm{Beta}(a{+}n_1,\ b{+}n_0)$ is a **complete answer**:

- **Point estimates** — posterior mean $\mathbb{E}[\theta \mid \mathcal{D}] = \dfrac{a + n_1}{a + b + N}$, or the mode (MAP).
- **Intervals** — central 95% credible interval, read straight off the density.
- **Shrinkage, made visible** — the mean is a *weighted blend* of prior mean and sample frequency:

$$\frac{a + n_1}{a + b + N} = \underbrace{\frac{a + b}{a + b + N}}_{\text{prior weight}} \cdot \frac{a}{a + b} + \underbrace{\frac{N}{a + b + N}}_{\text{data weight}} \cdot \frac{n_1}{N}$$

As N grows, the data weight $\to 1$ — the sensitivity figure, as algebra.

Predicting the next patient

For the **next** patient, average the Bernoulli model over the posterior — the **posterior predictive**:

$$p(y_{\text{new}} = 1 \mid \mathcal{D}) = \int_0^1 \theta \, p(\theta \mid \mathcal{D}) \, d\theta = \mathbb{E}[\theta \mid \mathcal{D}] = \frac{a + n_1}{a + b + N}$$

- Prediction **integrates over parameter uncertainty** instead of plugging in a point estimate.
- With $a = b = 1$: $\frac{n_1+1}{N+2}$ — Laplace's *rule of succession*; never exactly 0 or 1 from finite data.

Recap & bridge

- **Bayes**: posterior $\propto$ likelihood $\times$ prior — belief, updated.
- **Beta-Binomial conjugacy**: update = **addition** — $\mathrm{Beta}(a, b) \to \mathrm{Beta}(a+n_1, b+n_0)$.
- **Posterior** = point estimates + intervals + predictions, from one object.

Next (Lecture B): the same conjugate trick on the *regression weights* — Gaussian prior + Gaussian likelihood $\to$ Gaussian posterior for $\mathbf{w}$. That closed form is where **Part 1 ends**.